# Discrete-Time Markov Chains (DTMCs): Part 2

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# References


- S.M. Ross, “Stochastic Processes”: Chapter 4
- R. Wolff: Chapter 3
- S.M. Ross, “Introduction to Probability Models”: Chapter 4
- M. Harchol-Balter: Chapters 8 and 9
- A. Kumar, “Discrete Event Stochastic Processes”: Chapter 2

# Recall: Probability Vectors and Transition Probability Matrices

- For $j \geq 0$:
  - Let $P(X_0 = j) = a_j$
  - For $n \geq 1$, let $P(X_n = j) = a_j^{(n)}$
- Consider the following row vectors:
  - $a = [a_0, a_1, \dots]$
  - $a^{(n)} = \left[a_0^{(n)}, a_1^{(n)}, \dots\right]$
- $a$ and $a^{(n)}$ are called *probability vectors* since their elements are non-negative and sum to $1$
- Suppose transition probability matrix is $P$
- $a^{(n)}$:
  - $a^{(n-1)}P$
  - $aP^n$

# Example

- Suppose $X_n$ is brand chosen on a consumer's $n$th purchase:
  - ❑ modeled as a DTMC
- Market share of brand $j$:
  - ❑ fraction of all purchases that are of brand $j$

$$1) \quad \lim_{n\to\infty} \frac{\sum_{s=1}^{n} 1_{\{X_s=j\}}}{n}$$

- Expectation of limit in 1):

$$2) \quad \lim_{n\to\infty} \frac{\sum_{s=1}^{n} a_j^{(s)}}{n}$$

- Alternative definition of market share:

$$3) \quad \lim_{n\to\infty} a_j^{(n)} = \pi_j, \text{ say,}$$

  - ❑ where $\pi_j \geq 0$, for all $j$, and $\sum_j \pi_j = 1$

- We have assumed that for each $j$, the limit in 3) exists and is independent of initial vector $a$
- Recall:

$$4) \quad a^{(n)} = a^{(n-1)}P$$

- Taking limits as $n \to \infty$ on both sides of 4), we get:

$$5) \quad \pi = \pi P$$

- Also, by 3), it follows that:
  - ❑ the limit in 2) equals $\pi_j$
- Thus, market share of brand $j$ can be found by solving the system of equations 5) along with $\sum_j \pi_j = 1$

# Example (contd.)

- Recall:

  *1)* $\lim_{n\to\infty} \frac{\sum_{s=1}^{n} 1_{\{X_s=j\}}}{n}$

  2) $\lim_{n\to\infty} \frac{\sum_{s=1}^{n} a_j^{(s)}}{n}$

  *3)* $\lim_{n\to\infty} a_j^{(n)} = \pi_j$

  *4)* $a^{(n)} = a^{(n-1)}P$

  *5)* $\pi = \pi P$

- It is possible that limit in 3) does not exist, but limit in 2) can be found by solving the system of equations 5)
- E.g., let:
  - ❑ $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, and
  - ❑ $a = [a_0, a_1]$, where $a_0 + a_1 = 1$
- $a^{(n)}$:
  - ❑ $[a_0, a_1]$ for $n$ even
  - ❑ $[a_1, a_0]$ for $n$ odd
- Limit in 2):

  *6)* $\pi_0 = \pi_1 = \frac{1}{2}$

- The vector in 6) is also the unique solution of 5) that is a probability vector
- Limit in 3):
  - ❑ does not exist, unless we choose $a = \left[\frac{1}{2}, \frac{1}{2}\right]$

# Positive Recurrence of Irreducible DTMCs

- Recall:
  - **Theorem**: If $i \leftrightarrow j$, then:

    *a)* $P\left\{\lim_{t\to\infty} \frac{N_j(t)}{t} = \frac{1}{\mu_{jj}} \middle| X_0 = i\right\} = 1$

    *b)* $\lim_{t\to\infty} \frac{M_j(t)}{t} = \lim_{t\to\infty} \frac{\sum_{n=1}^{t} P_{ij}^n}{t} = \frac{1}{\mu_{jj}}$
- **Theorem**:
  - For an irreducible DTMC, all states are positive recurrent iff there is a unique stationary probability vector $\pi$ satisfying $\pi = \pi P$:
    - where $\pi_j = \frac{1}{\mu_{jj}} > 0$, for all $j \in \{0,1,\dots\}$
  - Otherwise, either all states are transient or all states are null recurrent:
    - In either case, no stationary probability vector exists
- From above two theorems, it follows that if an irreducible DTMC is positive recurrent, then fraction of time for which process is in state $j$ is:
  - $\pi_j = \frac{1}{\mu_{jj}}$, where $\pi_j$ is element $j$ of vector $\pi$, which is the unique stationary probability vector

# Example

- Recall the brand-switching example we discussed earlier:
  - $X_n$ is brand chosen on a consumer's $n$th purchase
  - DTMC is irreducible with transition probability matrix $P$
- Suppose there are a total of $c < \infty$ brands:
  - so, the DTMC is finite and hence positive recurrent
- Market share of brand $j$:
  - will be equal to $\pi_j$, element $j$ of vector $\pi$, which is the unique stationary probability vector
- First, consider symmetric case where:
  - $P_{jj} = \frac{1}{2}$, for all $j$,
  - $P_{ij} = \frac{1}{2(c-1)}$, for all $i \neq j$
- **Exercise**: Show that $\pi = \left[\frac{1}{c}, \frac{1}{c}, \dots, \frac{1}{c}\right]$ satisfies $\pi = \pi P$

# Example (contd.)

- Now, suppose manufacturer of brand $x$ wishes to increase that brand's market share by increasing $P_{xx}$
- Suppose:
  - ❑ $P_{xj} = \frac{1-P_{xx}}{c-1}$ for all $j \neq x$
- For all $i \neq x$, assume that as before:
  - ❑ $P_{ii} = \frac{1}{2}$,
  - ❑ $P_{ij} = \frac{1}{2(c-1)}$, for all $j \neq i$
- Consider a new two-state DTMC, where:
  - ❑ state $0$ denotes brand $x$, and
  - ❑ state $1$ denotes *not* brand $x$
- This is a two-state DTMC with transition probability matrix:
  - ❑ $P = \begin{bmatrix} P_{xx} & 1-P_{xx} \\ \frac{1}{2(c-1)} & \frac{(2c-3)}{2(c-1)} \end{bmatrix}$
- Solution of $\pi = \pi P$ along with $\pi_0 + \pi_1 = 1$ gives the market share of brand $x$ as:
  - ❑$\pi_0 = \frac{1}{(2c-2)(1-P_{xx})+1}$
- $\pi_0$:
  - ❑equals $\frac{1}{c}$ as before if $P_{xx} = \frac{1}{2}$
  - ❑increases in $P_{xx}$
  - ❑$\rightarrow 1$ as $P_{xx} \rightarrow 1$
- **Note**:
  - ❑Usually, combining states as we have done destroys Markovian property, i.e., we would no longer have a DTMC
  - ❑However, this technique works here because of the specific symmetrical structure of the state transition diagram

# Example

- Consider a DTMC with an infinite number of states
- Let state space be $\{0,1,2,\dots\}$ and transition probability matrix be:
  - $P = \begin{bmatrix} 0 & 1 & 0 & \dots & & \\ q & 0 & p & 0 & \dots & \\ 0 & q & 0 & p & 0 & \dots \\ & \vdots & & \vdots & & \vdots \end{bmatrix}$,
  - where $qp > 0$ and $q + p = 1$
- Is this DTMC irreducible?
  - Yes
- System of equations $\pi = \pi P$ becomes:
  - $\pi_0 = q\pi_1$,
  - $\pi_1 = \pi_0 + q\pi_2$,
  - $\pi_j = p\pi_{j-1} + q\pi_{j+1}, j \in \{2,3,\dots\}$
- Solving this system recursively, we get:
  - $\pi_1 = \frac{\pi_0}{q}$,
  - $\pi_{j+1} = \frac{p\pi_j}{q}, j \in \{1,2,\dots\}$
- $\pi_j$ in terms of $\pi_0$:
  - $\pi_j = \left(\frac{p}{q}\right)^{j-1} \frac{\pi_0}{q}, j \in \{1,2,\dots\}$
- $\sum_{j=0}^{\infty} \pi_j$:
  - *1)* $\pi_0 \left[1 + \sum_{j=1}^{\infty} \frac{1}{q} \left(\frac{p}{q}\right)^{j-1}\right]$
- The infinite sum in 1) converges iff $p < q$
- Hence, when $p < q$, all states are positive recurrent, and stationary distribution given by:
  - $\pi_0 = \frac{(q-p)}{2q}$
  - $\pi_j = \left(\frac{p}{q}\right)^{j-1} \frac{(q-p)}{2q^2}, j \in \{1,2,\dots\}$
- Can be shown that:
  - all states are transient if $p > q$
  - all states are null recurrent if $p = q$

# Positive Recurrence of Irreducible DTMCs (contd.)

- Recall:
  - **Theorem**:
    - For an irreducible DTMC, all states are positive recurrent iff there is a unique stationary probability vector $\pi$ satisfying $\pi = \pi P$:
      - where $\pi_j = \frac{1}{\mu_{jj}} > 0$, for all $j \in \{0,1, \dots\}$
    - Otherwise, either all states are transient or all states are null recurrent:
      - In either case, no stationary probability vector exists
- Suppose:
  - $P$ is the transition probability matrix of an irreducible DTMC
  - $\pi$ is a vector with non-negative elements such that:
    - $\pi = \pi P$, and
    - $\pi_i > 0$, for *some* $i \in \{0,1, \dots\}$
- Consider any $j \in \{0,1, \dots\}$
- Since the DTMC is irreducible, there exists $m \geq 1$, s.t. $P_{ij}^m > 0$
- Also, $\pi = \pi P^m$, and hence:
  - $\pi_j = \sum_{l=0}^{\infty} \pi_l P_{lj}^m > 0$
- Hence, *every* element of $\pi$ is positive
- Also, note that for every $\alpha > 0$:
  - $(\alpha\pi) = (\alpha\pi)P$
- Hence, $\pi$ can be normalized so that its elements constitute a PMF on $\{0,1, \dots\}$